

Ketan Kumar Bagewadi

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PROFESSIONAL SUMMARY

Software Engineer specializing in **Python, Machine Learning, LLMs, NLP, RAG and Agentic AI (LangGraph, MCP)**. Builds and deploys scalable **AI applications** and backend systems with strong hands-on experience in **Docker, Kubernetes, Jenkins**, and **cloud technologies**. Passionate about developing production-ready AI solutions and seeking challenging opportunities in AI/ML or DevOps roles.

EXPERIENCE

NetSense CyberSecurity Pvt. Ltd.

Jun 2026 – Present

- Software Engineer skilled in **Python, Machine Learning**, and **LLM**, building **AI-driven** backend systems and security analytics tools.
- Built a 12-rule network anomaly detection engine (beaconing, port scans, SYN floods, DNS tunneling, exfiltration) using GPU-accelerated (**cuDF**) processing of NetFlow/DNS logs across multiple network segments.
- Integrated a local **Llama 3.1 8B** model to auto-triage flagged hosts, producing evidence-backed verdicts (true/false positive) with MITRE ATT&CK mapping — cutting manual SOC review time.
- Optimized the phishing and malicious website detection system to run efficiently on very low resources, making it production-ready for high-scale deployment on gateways and edge environments.

Smile Security and Surveillance Pvt. Ltd (NetSense CyberSecurity Pvt. Ltd.)

Mar 2025 – May 2026

- Software Developer working on **Python, Machine Learning** and **Deep Learning** to detect and flag **phishing and malicious websites**. Achieved 95%-97% accuracy in detecting true phishing websites on live traffic.
- Designed and deployed dual production website classification for **40+ categories** using **Llama 3.1 8B** and 250-thread **Selenium** pipeline and also implemented using the **Sentence-t5-xxl** NLP model with cosine-similarity and zero-shot classification.
- Deployed **CRM** system using **Python, Django, Django REST Framework, JavaScript, Ajax, Linux, Nginx, Azure Directory, DigitalOcean Droplet** and **PostgreSQL**.
- Optimised **CRM Web App** to query from and sync with **2+ Database servers**. Accomplished **30%** faster read and write operations.

PROJECTS

LLM-Powered RAG ChatBot | Python, LangChain, OpenAI, Ollama, LLM, RAG

[Github Link](#)

- Developed an AI chatbot using **GPT-4o-mini**, **Llama 3** fallback, and **LangChain** to deliver reliable, real-time conversational responses.
- Built a **RAG-based chatbot** using **ChromaDB**, **document chunking**, **embeddings**, and **semantic retrieval** for context-aware responses and improved answer accuracy. **Containerized** the frontend and backend using **Docker**, **orchestrated** services with **Docker Compose**, and deployed the RAG application using **Kubernetes**.
- Implemented document upload (PDF/TXT/DOCX/MD), streaming responses via Server-Sent Events, automatic fallback between **OpenAI** and local **Ollama**, and source citation in answers using **FastAPI** + Tailwind frontend.

Multi-Agent Research Assistant | Python, LangGraph, MCP, Claude API, Ollama, FastAPI

[Github Link](#)

- Built a **multi-agent** research assistant with **LangGraph** — Planner, Searcher, Summarizer, Verifier, and Writer agents working together with tool calling and multi-step reasoning.
- Integrated **MCP (Model Context Protocol)** servers wrapping **Tavily search** for retrieval and **fact-checking**, with a self-correcting verifier–retry loop to reduce hallucinations and ground outputs in real sources.
- Built a provider-agnostic **LLM backend** (Claude/OpenAI/Groq/Ollama) using **FastAPI**, with real-time **SSE** streaming and **SQLite** storage.

DeepFake Detection | Python, Google Colab, PyTorch, HTML, CSS, JavaScript

[Github Link](#)

- Developed a **deepfake detection** project utilizing **deep learning** techniques. Employed the **FaceForensics++** and **Celeb-DF** database to train a robust **machine learning** model.

- Implemented an architecture combining **ResNeXt** and **LSTM** models for accurate identification and classification of deepfake videos.
- Built a full **Django** web application with video upload interface, real-time inference, **Docker** support (including **GPU**), and **Nginx** configuration for production deployment.

AI ChatBot Android App | Android Studio, Java, Firebase, Google PaLM-2 API

[Github Link](#)

- Developed an **Android application** proficient in processing text input and generating responses for a wide array of questions sourced from the **Google PaLM-2 API**.
- Engineered an application with capabilities of **ChatGPT** like conversational capabilities.
- Integrated **Firebase Authentication** for user sign-up and login, gating access to the chat screen behind verified credentials.

TECHNICAL SKILLS

Languages: Python, C | **Frameworks & Libraries:** PyTorch, Scikit-learn, Pandas, Django REST Framework, Bootstrap, TailwindCSS | **Databases:** PostgreSQL, MySQL | **DevOps:** Docker, Kubernetes, Jenkins, Nginx, DigitalOcean | **Cloud:** AWS, Azure | **Version Control:** Git, GitHub | **Other:** Unix/Linux, Machine Learning, Deep Learning, NLP, Data Analysis, Web Scraping, LLM, Bert, RAG, Llama, Prompt Engineering, AI Tools (ChatGPT, Claude AI, Grok AI, Google AI, Cursor, Mistral AI)

EDUCATION

Bachelor of Engineering in Computer Science & Engineering

KLE Dr. M.S. Sheshgiri College of Engineering and Technology, Belgaum, Karnataka

June 2024

CERTIFICATIONS

AWS Cloud Practitioner Essentials Course (**Amazon**), Introduction to Python (**Sololearn**, **Infosys Springboard**), Introduction and Intermediate SQL (**Great Learning**), Generative AI (**GeeksforGeeks**), Introduction to Data Science (**Cisco**)